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A															
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C															
D															
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F															
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM DRC BUS PT H05															
				Date Prep. 15.11.2018		PROJECT TITLE				ABB EPMV		COVER SHEET		Scale	
				Prepared YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-01 20kV MAIN SUBSTATION		Customer PT Multimas Nabati Asahan	
00		Issued for Approval		28.12.18 YR		Checked AW		SCHEMATIC DIAGRAM				Drawing Nbr.		IDUNISEC-BE318042-301	
Ind.		Revision		Date		Name		Approved AI		Based on: BE318042				Customer Doc. Nbr.	
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F					Date Prep.	15.11.2018	PROJECT TITLE		ABB EPMV	Index of Sheet	Scale		= H05	F
					Prepared	YR	MV Switchgear for MNA Serang Plant			MVD-01 20kV MAIN SUBSTATION	Customer	PT Multimas Nabati Asahan	& 308	
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Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				DRC BUS PT	Customer Doc. Nbr.		Cont.	16
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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS				-BGB5	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED		
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)				-BGB6	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY		
B	DESIGNATION	DESCRIPTION			-BGB7	POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR		
	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)			-BGB8...-BGB10	POSITION SWITCHES OF CIRCUIT-BREAKER		
C	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1			-BGB11	POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING		
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2			-BGB12	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION		
D	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3			-BGB13	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION		
	-BAR	VOLTAGE PROTECTION RELAY			-BGD	POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.		
E	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1			-BGD1	POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR		
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2			-BGD2	POSITION SWITCH OF CABLE COMPARTMENT DOOR		
F	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3			-BGE1	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION		
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1			-BGE2	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION		
A	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2			-BGE3	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		
	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3			-BGE4	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR		
B	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1			-BGE5	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR		
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2			-BGE7	POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION		
C	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3			-BGE8	POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS		
	-BCD	DIFFERENTIAL PROTECTIVE RELAY			-BGE11	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION		
D	-BCF	FEEDER PROTECTION RELAY			-BGE12	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION		
	-BCG	GENERATOR PROTECTION RELAY			-BGE13	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		
E	-BCM	MOTOR PROTECTION RELAY			-BGE14, -BGE15	POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR		
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER			-BGF	POSITION SWITCHES OF MEDIUM VOLTAGE FUSES		
F	-BCP	TRANSFORMER PROTECTION RELAY			-BG11	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION		
	-BCR	CURRENT PROTECTION RELAY			-BG12	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION		
A	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1			-BG13	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2			-BG14	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR		
B	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3			-BG15	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR		
	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1			-BG16	POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION		
C	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2			-BG17	POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION		
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3			-BG18	POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS		
D	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1			-BG19	POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR		
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2			-BG111	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION		
E	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3			-BG112	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION		
	-BCZ	DISTANCE PROTECTION RELAY			-BG113	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		
F	-BEF	FREQUENCY PROTECTION RELAY			-BG114, -BG115	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR		
	-BER	SUPERVISION RELAYS			-BG121	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION		
A	-BES	SYNCHRONIZING RELAY			-BG122	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION		
	-BET	THERMAL PROTECTION RELAY			-BG123	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		
B	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT			-BG124, -BG125	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR		
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)						
C	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT						
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)						
D	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER						
	-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS						
E								
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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR						
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR						
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.						
B	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT						B
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE						
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT						
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT						
C	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT						C
	-BGK		POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET						
	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST						
	-BGS1...-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.						
D	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE						D
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE						
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1						
	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2						
E	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3						E
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS						
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1						
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2						
F	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						F
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER						
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS						
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1						
	-BGV2		POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2						
	-BM		HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3						
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES						
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS						
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR						
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR						
	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK						
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE						
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE						
	-BR		FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS						
	-BT		THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS						
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS						
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY						
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS						
	-BU		BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS						
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS						
	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)						
	-BZ,-BZ4		MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS						
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES						
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS						
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS						
									-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						
									-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD						
									-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1						
									-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2						
									-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS						
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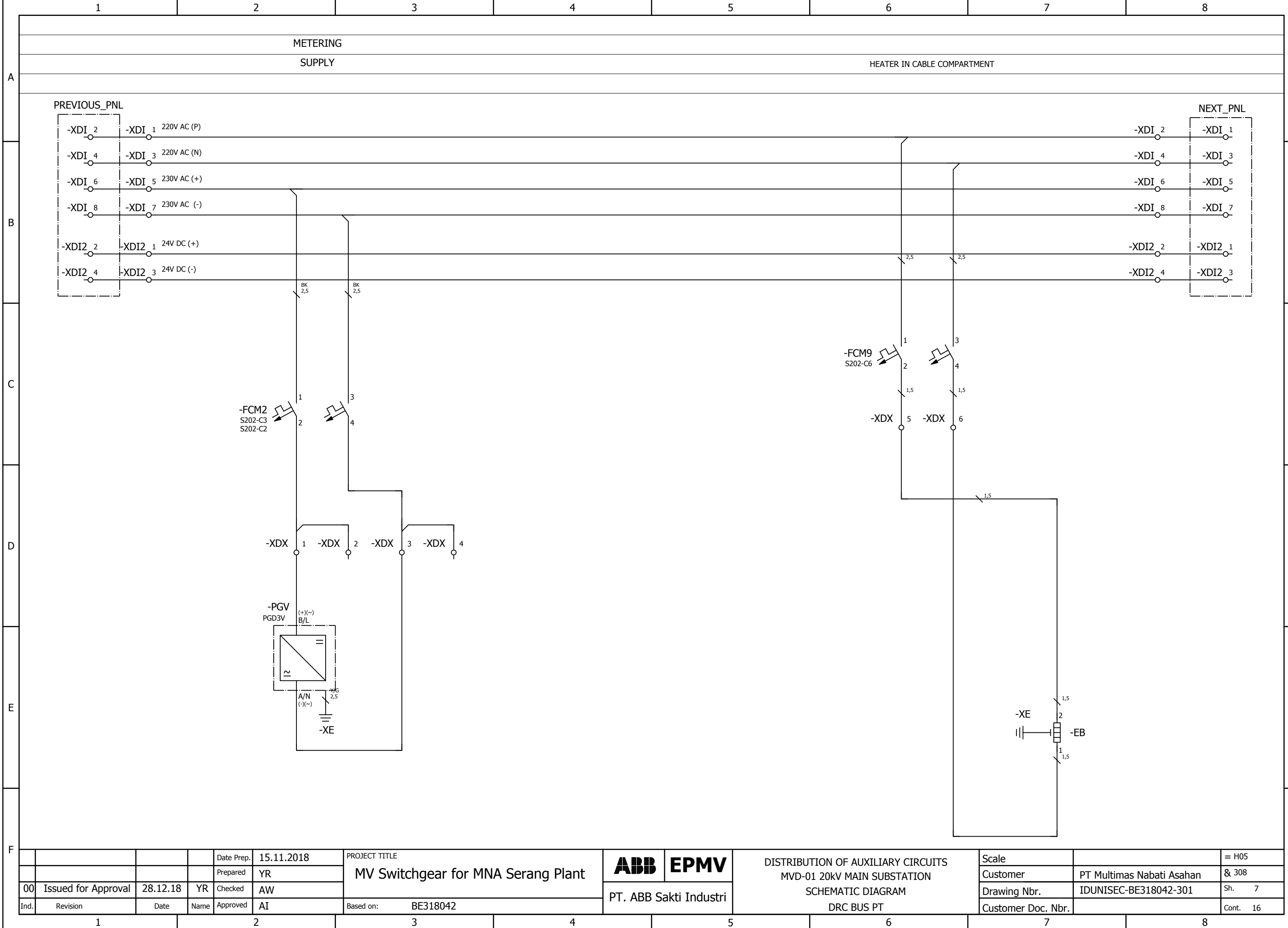
A	1	2			3			4			5		6		7		8																	
	-MAE		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE															-QBH		MANUAL CIRCUIT-BREAKERS														
	-MAE1		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2															-QBL		LINKS														
	-MAE8		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS															-QBM		MINIATURE SWITCH-DISCONNECTORS														
																		-QBS		SWITCH-DISCONNECTORS														
																		-QBT		TRUCK														
																		-QCA		EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING														
																		-QCE		EARTHING SWITCH														
																		-QCE1		EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)														
B	-MBO1		FIRST OPENING RELEASE OF CIRCUIT-BREAKER															-QCE2		EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)														
	-MBO2		SECOND OPENING RELEASE OF CIRCUIT-BREAKER															-QQ		CLUTCHES														
	-MBO3		OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER															-RAA		ANTIFERRORESONANCE RESISTOR														
	-MBO4		OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS															-RAD		DIODES														
	-MBU		UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER															-RAI		INDUCTORS														
	-PFB		BLUE SIGNAL LAMPS															-RAR		RESISTORS														
	-PFF		FLAG RELAYS															-RF		FILTERS														
	-PFG		GREEN SIGNAL LAMPS															-RLE1		ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING														
	-PFR		RED SIGNAL LAMPS															-RLE2		ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT														
	-PFS		SHORT CIRCUIT INDICATORS															-RLE3		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE														
C	-PFV		VOLTAGE INDICATORS																															
	-PFV1		VOLTAGE INDICATORS ON FEEDER SIDE															-RLE4		ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION														
	-PFV2		VOLTAGE INDICATORS ON BUSBAR SIDE															-RLE5		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD														
	-PFW		WHITE SIGNAL LAMPS															-RLE6		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1														
	-PFX		CROSS INDICATORS, ELECTROMECHANICAL INDICATORS																															
	-PFY		YELLOW SIGNAL LAMPS															-RLE7		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2														
	-PGA		AMMETERS															-RLE8		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2														
	-PGC		COUNTERS																															
	-PGF		FREQUENCYMETERS															-RLE9		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS														
	-PGH		HOURMETERS															-RLE11		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4														
D	-PGI		PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE															-RLE12		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5														
	-PGI1		HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)																															
	-PGI2		HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)																															
	-PGJ		ACTIVE ENERGY METERS															-SFA		AMMETRIC SWITCHES														
	-PGK		REACTIVE ENERGY METERS															-SFC		CONTROL SWITCHES, CLOSING PUSH-BUTTONS														
	-PGM		MULTIFUNCTION INDICATORS															-SFL		LOCKING CONTACTS														
	-PGP		POWER-FACTOR METERS															-SFO		OPENING PUSH-BUTTONS														
	-PGQ		VARMETERS															-SFR		RESET PUSH-BUTTONS														
	-PGS		SYNCHRONOSCOPES															-SFS		SELECTOR SWITCHES														
	-PGV		VOLTMETERS															-SFT		TEST PUSH-BUTTONS														
E	-PGW		WATTMETERS															-SFU		UNLOCKING PUSH-BUTTONS														
	-PJ		ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)															-SFV		VOLTMETRIC SWITCHES														
	-QAB		CIRCUIT-BREAKERS															-TA		POWER TRANSFORMERS														
	-QAC		CONTACTORS (FOR POWER)															-TB		CONVERTER														
	-QBD		DISCONNECTORS															-TB1		RECTIFIER FOR FIRST OPENING RELEASE														
	-QBD1		FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)															-TB2		RECTIFIER FOR SECOND OPENING RELEASE														
	-QBD2		REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)															-TB3		RECTIFIER FOR CLOSING RELEASE														
	-QBD3		FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)															-TB4		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING														
	-QBD4		TWO-WAY DISCONNECTOR															-TB5		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT														
	-QBD5		TWO-WAY DISCONNECTOR ON BUSBAR RISER															-TB6		RECTIFIER FOR UNDERVOLTAGE RELEASE														
F																		-TB7		ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER														

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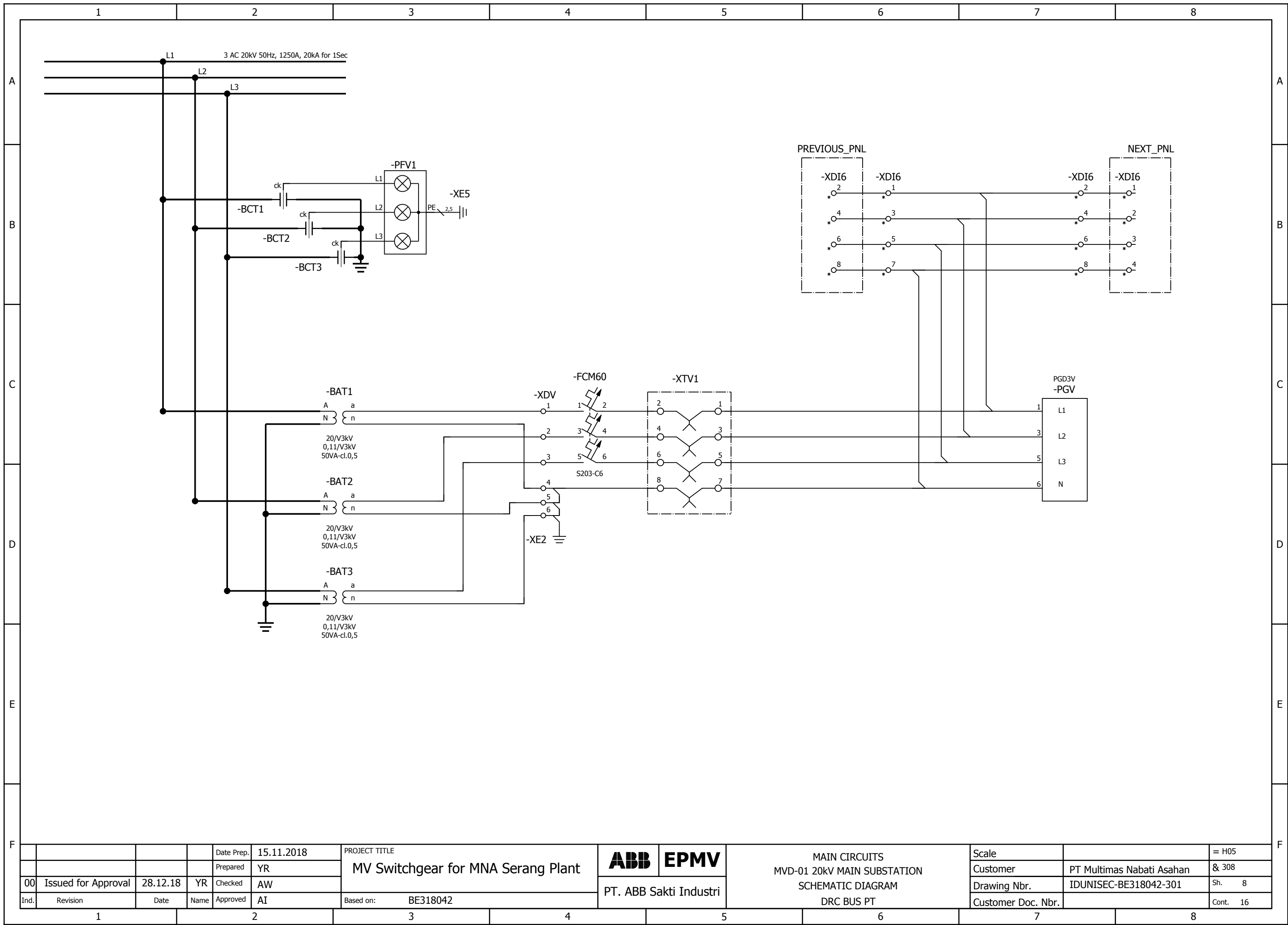
A	-TFA		ACTIVE POWER TRANSDUCERS		-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS		A
	-TFC		CURRENT TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING		
	-TFF		FREQUENCY TRANSDUCERS		-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION		
	-TFJ		ACTIVE ENERGY TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS		
	-TFK		REACTIVE ENERGY TRANSDUCERS		-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS		
	-TFM		MULTIFUNCTION TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS		
	-TFP		POWER-FACTOR TRANSDUCERS		-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS		B
	-TFQ		REACTIVE POWER TRANSDUCERS			TERMINAL BLOCK FOR SPECIAL USE		
	-TFS		SIGNAL CONVERTERS		-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD		
	-TFV		VOLTAGE TRANSDUCERS			TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS		
B	-WA		BUSBARS		-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT		
	-WBC		POWER CABLES			CURRENT SOCKETS		
	-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.		-XDS	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB		
	-WGC		CONTROL CABLES			CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK		
	-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)		-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK		
	-WH		OPTICAL FIBRES			TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS		
	-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS		-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1		
	-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2		
	-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2		-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2		
	-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3		
	-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-XDV3	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR		
	-XDB		CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS			CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR		
	-XDB1		CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		-XDX	SUPPORT TERMINAL BLOCK		
	-XDB2		CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE		
	-XDB10...-XDB89		CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE		
	-XDB7		CONNECTOR FOR TRUCK POSITION CONTACTS			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE		
	-XDB91		CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE		
	-XDB92		CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE		
	-XDB93		CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR		-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE		
	-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT		
D	-XDC		CUSTOMER TERMINAL BLOCK		-XE8			
	-XDC1		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER					
	-XDC2		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR		-XE9			
	-XDC3		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT					
	-XDC4		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS					
	-XDC5		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER					
	-XDE		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE					
	-XDE1		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE					
	-XDE2		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2					
	-XDE3		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2					
	-XDF		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS					
	-XDF1		CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR					
	-XDF2		CONNECTOR FOR REAR VENTILATOR CIRCUITS					
	-XDF3		CONNECTOR FOR FRONT VENTILATOR CIRCUITS					
	-XDH		TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)					
	-XDI		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)					

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM	Scale		= H05
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 308
00	Issued for Approval	28.12.18	YR	Checked	AW			PT. ABB Sakti Industri		Drawing Nbr.	IDUNISEC-BE318042-301	Sh.	6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				Customer Doc. Nbr.		Cont.	16

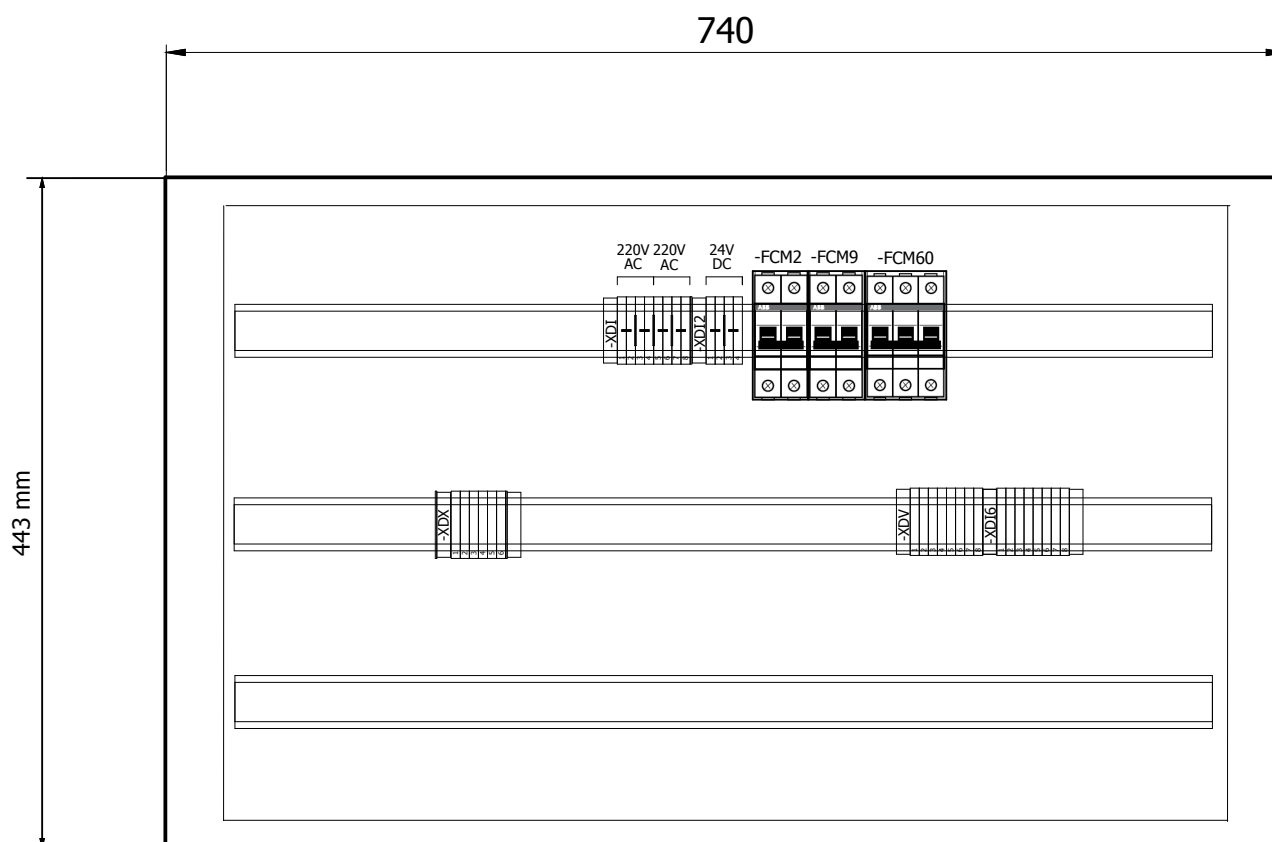
1		2		3		4		5		6		7		8	
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	LV BOX LAYOUT MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05
				Prepared	YR				Customer	PT Multimas Nabati Asahan	& 308
00	Issued for Approval	28.12.18	YR	Checked	AW				Drawing Nbr.	IDUNISEC-BE318042-301	Sh. 10
Ind.	Revision	Date	Name	Approved	AI	Customer Doc. Nbr.			Cont. 16		
Based on: BE318042						PT. ABB Sakti Industri					

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	1	2	3	4	5	6	7	8
A	Summarized parts list							
	IDABB_F02_005-IDABB							
	Carry over: 0.00							
	TAG ID	DESCRIPTION	MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS		
	-EB	HEATER 230VAC, 50W	SANYANG	CDXN010033P0009	1	/7.7:E		
	-FCM2	MCB S202-C2 MCB. 2P, 2A	ABB	2CDS252001R0024	1	/7.2:C		
	-FCM9	MCB S202-C6 MCB. 2P, 6A	ABB	2CDS252001R0064	1	/7.6:C		
	-FCM60	MCB S203-C6 MCB. 3P, 6A	ABB	2CDS253001R0064	1	/8.4:C		
	-PGV	Digital AC Voltmeter Model : DPM PGD3V, 48 X 96 MM	RISHABH	DPM PGD3V	1	/7.2:D		
	-XTV1	Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting	YONGSUNG	YSPTT-04C	1	/8.5:C		
B								
C								
D								
E								
F								

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